



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Simple Mobile App To Support Farmers

Semester: Fall 25-26

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1. PROJECT PROPOSAL

1.1 Background to the Problem

The Farmer Support App is a simple yet powerful mobile and web-based platform designed to assist farmers in managing agricultural activities more efficiently through digital technology. The main goal of this application is to provide farmers with easy access to essential information, market updates, expert advice, and weather forecasts — all in one integrated system. By digitizing agricultural support services, the app aims to reduce dependency on intermediaries, improve productivity, and ensure farmers receive fair value for their products.

The app will feature modules such as crop management guidance, real-time weather updates, market price monitoring, and direct buyer-seller connections. Farmers can create personal profiles to track their crops, receive notifications about pest control measures, and view up-to-date price trends in local and national markets. Buyers and suppliers can also register on the platform to post offers, negotiate prices, and communicate directly with farmers. By bridging the information and communication gap between stakeholders, the Farmer Support App enhances decision-making and supports sustainable agricultural development.

Overall, the project demonstrates how digital tools can empower rural communities and modernize traditional farming practices in an accessible, user-friendly, and cost-effective manner.

1.2 Selection of Process Model

Extreme Programming (XP) is selected for the development of the Farmer Support App because it emphasizes frequent releases, continuous testing, simplicity, and close communication with stakeholders. XP focuses on producing high-quality software while adapting quickly to changing requirements.

XP is ideal for projects with complex functionality and diverse user roles, as it allows the team to respond immediately to feedback and improve the system continuously.

Key reasons for choosing XP:

- **Continuous Feedback:** Code is tested and reviewed frequently, reducing bugs and improving reliability.
- **Short Development Cycles:** Features are delivered in small increments, so working software is always available.

- Customer Involvement: User feedback is continuously incorporated to improve usability.
- Simplicity: The code is kept simple and easy to modify for future requirements.

1.2.1 Alignment of XP with the Farmer Support App

The Extreme Programming model aligns closely with the requirements and goals described in the Software Requirements Specification of the Farmer Support App.

1.2.2 Modular and Incremental Development

The Software Requirements Specification describes the app as a modular system with independent features such as user management, weather information, marketplace services, and communication tools.

In XP:

- Each feature is developed incrementally and tested continuously.
- Essential features like user registration, login, and weather updates can be implemented first.
- Advanced features, such as expert advice, supplier advertisements, and alerts, can be added later without affecting existing functionality.

This incremental approach ensures the system remains stable while new features are introduced.

1.2.3 Focus on User-Centric Design

XP emphasizes close collaboration with users. The Farmer Support App is intended for rural farmers who may not be tech-savvy.

- Early prototypes are released to gather user feedback.
- The interface and features are improved based on actual usage.
- This ensures the system is easy to use and meets the real needs of farmers, buyers, experts, and suppliers.

1.2.4 Supporting Multiple User Roles

The system includes five user roles: Farmers, Buyers, Experts, Suppliers, and Administrators.

- In XP, developers work with user stories to understand each role's requirements.

- Features for different roles are implemented incrementally.
- Frequent testing ensures all user roles function correctly and interact properly with the system.

1.2.5 Continuous Testing and Risk Reduction

XP practices include test-driven development (TDD) and frequent integration.

- Each new feature is tested immediately.
- Problems are detected and fixed early, reducing the risk of major system failures.
- High-priority features are implemented first to ensure the core functionality of the system is always working.

1.2.6 Extreme Programming (XP) Model — Description

Extreme Programming (XP) is an agile software development methodology that emphasizes flexibility, simplicity, continuous improvement, and close collaboration with users. XP is particularly effective for projects where requirements may change frequently, and rapid delivery of working software is essential.

XP focuses on building software through short development iterations, where features are designed, implemented, and tested continuously. The model encourages frequent releases so that users can regularly review system functionality and provide feedback. This helps the development team refine features early and avoid major rework later.

Key principles of the XP model include:

- Communication – Developers, customers, and stakeholders work closely throughout the project.
- Simplicity – The system is designed with only the necessary functionality, reducing complexity.
- Feedback – Continuous testing and user input help improve quality and usability.
- Courage & Adaptability – The team can easily modify or refactor code when requirements change.
- Respect – Collaboration and shared responsibility strengthen teamwork and productivity.

1.2.7 Extreme Programming (XP) Model Diagram

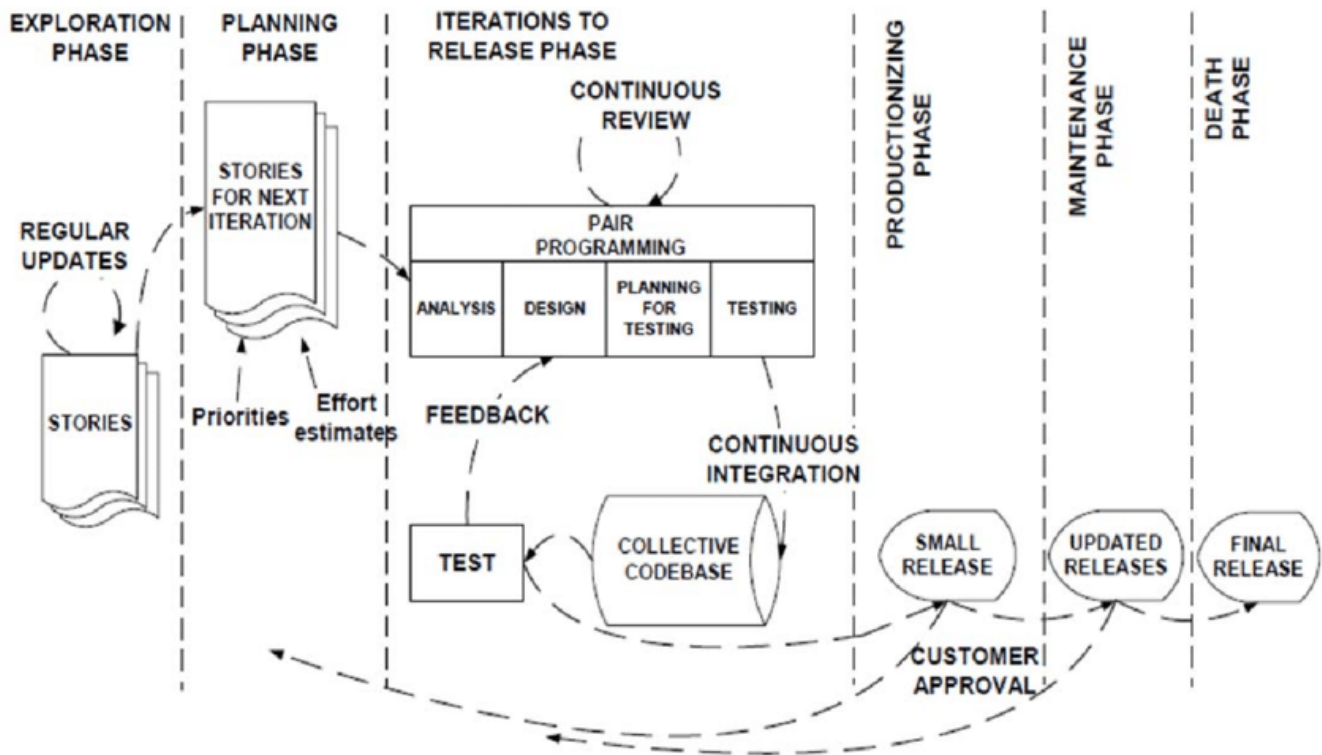


Fig 1: Shows Extreme Programming (XP) process model

Extreme Programming (XP) is well-suited for the Farmer Support App because it supports incremental feature delivery, continuous testing, and frequent feedback. XP ensures the system remains flexible, easy to maintain, and aligned with user needs.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

2.1 Scopes and Features

Users:

- Admin
- Buyers and Traders
- Agricultural Experts
- Suppliers

Features:

1. Profile Management

- Farmers, buyers, experts, and suppliers can register and create profiles with valid information.
- All users can log in securely and manage their personal profiles.

2. User Management

- Admin can control access of all users.
- Admin can manage all users(eg; remove, edit information.)

3. Information and Advisory Services

- Farmers can see weather forecasts, crop management tips, and pest control advice.
- Agricultural experts can share best practices and answer farmers' queries through chat or discussion forums.

4. Market and Trade Management

- Farmers can list their products and view current market prices.
- Buyers can browse available products, compare prices, and communicate directly with farmers for transparent deals.
- Suppliers can advertise their products such as seeds, fertilizers, and equipment for farmers' convenience.

5. Notification and Updates

- Farmers will receive timely notifications about weather alerts, crop guidance, and market changes.
- Buyers will receive timely notification if new crops are available in the system.
- Suppliers will receive notification for order of their products.

2.2 User Story Table

As A/An	I want to	So that	Acceptance Criteria
Admin	register and create my profile with valid information	I can enroll myself in the system and use the app securely	The system takes required fields (name, email, phone, password), validates them, creates an admin profile, and shows a success message after saving the data.
	log in and manage my personal profile	my information stays correct and I can access admin features securely	The system authenticates my email/phone and password, allows access only on correct credentials, shows my profile details, and lets me edit and save updated information.
	Control access permissions for all users	The system security is maintained and unauthorized access is prevented	System shows access control dashboard with role-based permissions
	View all registered users in the system	I can monitor user activities and manage	User list table displays all users with their profile details and registration dates

		accounts	
	Edit user information (profile details)	User data remains accurate and up-to-date	Edit form appears with existing user data, changes are saved successfully
	Remove users from the system	Inactive or unauthorized users cannot access the platform	User is deleted from database, confirmation popup appears before deletion
Farmer	Register and create a profile with my information	I can enroll myself in the system	Registration form appears with required fields, OTP sent to mobile, confirmation email sent to registered email
	Login with username and password	The system can authenticate me and I can trust it	After successful login, the home dashboard appears with my personal data and available features
	Manage my personal profile details	I can update my information anytime	Profile page shows current details with edit button, changes are saved successfully
	View weather forecasts for my region	I can plan my farming activities accordingly	Weather forecast page displays current and 7-day forecast with temperature, rainfall, and alerts
	Receive weather alerts and notifications	I can be prepared for extreme weather conditions	Push notifications appear for severe weather, SMS/email alerts sent to registered contact
	View crop management tips	I can improve my farming practices	Tips page displays seasonal crop guidance with images and step-by-step instructions
	Receive crop guidance alerts	I am informed about best practices for my crops	Notifications sent based on crop type and season, alerts appear in notification center

	View pest control advice	I can prevent crop damage	Pest control page displays common pests, symptoms, and recommended solutions
	List my agricultural products for sale	Buyers can see what I am producing	Product listing form appears, products added to marketplace with images and descriptions
	View current market prices for my crops	I can set competitive prices	Market price page displays real-time prices for different crops in local and national markets
	Browse and compare available products from suppliers	I can purchase quality seeds and fertilizers	Product catalog appears with filters, comparison feature shows price and features side-by-side
	Receive notifications about new available products	I am aware of new supplies in the market	Push notifications sent when new products listed, notification center shows product details
	Communicate directly with buyers about my products	I can negotiate fair prices without intermediaries	Chat interface appears with buyer, message history maintained, price negotiation possible
	View my sales reports and transaction history	I can track my income and sales performance	Sales report page displays order history, earnings, charts showing sales trends
Buyer	Register and create a profile in the system	I can participate in the agricultural marketplace	Registration form appears with business details, OTP verification, confirmation email sent
	Login with username and password	The system can authenticate me securely	After successful login, dashboard appears with my account information and available products
	Manage my personal profile and business information	I can keep my profile updated	Profile page shows current information with edit button, changes saved successfully
	View available agricultural products listed by farmers	I can explore what is available in the market	Product catalog appears with filters by crop type, price range, and location
	Compare prices of products from different farmers	I can make informed purchasing decisions	Comparison feature shows multiple products side-by-side with prices and details
	Communicate directly with farmers about their products	I can negotiate prices and place orders	Chat interface appears with farmer, conversation history maintained, order discussion possible
	Receive notifications when new products are listed	I am aware of new agricultural products available	Push notifications sent when farmers list new products, notification center shows product details
	Place and receive orders from	I can purchase needed inputs like seeds and	Order form appears, order confirmation sent, order status tracking page available

	suppliers	fertilizers	
	Receive product order notifications from suppliers	I am aware that my ordered products are ready	Notifications sent when supplier processes order, order status changes in real-time
	Participate in discussion forums	I can share market insights and best practices	Forum page appears with ability to create topics, reply to discussions, and network with others
	View my purchase history and transaction reports	I can track my spending and business performance	Transaction report page displays order history, payments made, charts showing purchase trends
Supplier	Register and create a profile with my business information	I can list my products and reach farmers	Registration form appears with business details, OTP verification, confirmation email sent
	Login with username and password	The system can authenticate me securely	After successful login, dashboard appears with my account and product management options
	Manage my personal profile and business details	I can keep my business information current	Profile page shows current details with edit button, changes saved successfully
	Advertise my agricultural products (seeds, fertilizers, equipment)	Farmers can discover and purchase my products	Product listing form appears, products added to marketplace with images and descriptions
	Manage my product inventory	I can update stock levels and availability	Inventory management page displays all products with stock quantities, edit and update options
	Receive notifications when farmers/buyers place orders for my products	I am aware of incoming orders immediately	Push notifications sent for new orders, notification center shows order details and customer info
	Communicate directly with buyers about product orders	I can provide order updates and resolve issues	Chat interface appears with buyer, conversation history maintained, order discussion possible
	Process and manage customer orders	I can fulfill orders efficiently	Order management page displays pending orders with customer details, status update options
	Participate in discussion forums	I can promote my products and engage with the agricultural community	Forum page appears with ability to create topics, reply, and build reputation
	View my sales reports and transaction history	I can monitor my business performance and revenue	Sales report page displays order history, revenue earned, charts showing sales trends

Agricultural Expert	Register and create a profile with my expertise details	I can establish my credibility and reach farmers	Registration form appears with qualification details, OTP verification, confirmation email sent
	Login with username and password	The system can authenticate me securely	After successful login, dashboard appears with my profile and expert tools
	Manage my personal profile and expertise information	I can update my qualifications and specializations	Profile page shows current details with edit button, changes saved successfully
	Share best practices and farming knowledge	Farmers can learn from my expertise	Best practices page allows creating articles with text, images, and detailed instructions
	Answer farmers' queries through chat or direct messaging	I can provide timely expert advice	Chat interface appears when farmers request help, message notifications received, response tracking available
	Participate in discussion forums	I can engage with the farming community and answer public questions	Forum page appears with ability to create expert topics, answer queries, and build reputation
	View notifications about farmer inquiries	I am aware of new help requests from farmers	Push notifications sent for new inquiries, notification center shows farmer details and questions
	Communicate with farmers and other experts	I can collaborate and share knowledge	Chat and messaging interface available, can connect with multiple users simultaneously
	View my engagement statistics	I can track my contribution to the platform	Statistics page displays number of answered queries, forum contributions, and farmer feedback
	Participate in discussion forums and share best practices	I can help the entire farming community at once	Forum page displays my topics and responses, ability to edit and manage my contributions

Common User Stories

As A/An	I wan to	So That	Acceptance Criteria
All Users	Change my account password	My account remains secure and protected	Password change form appears, old password verified, new password set, confirmation message shown
	Request a new	I can regain access to	Password reset link sent to registered

	password if I forget mine	my account	email, OTP verification required, new password set successfully
	View my personal profile details	I can see all information I provided during registration	Profile page displays all registered information in organized format
	Update my personal information	My profile remains accurate and current	Edit profile form appears with existing data, changes saved successfully, confirmation message shown
	Logout from my account	I can ensure my account security on shared devices	Logout button appears in menu, session ends, redirected to login page
	Receive real-time notifications	I am informed about important system updates immediately	Notification center appears, push notifications sent, notification history maintained
	Search for content in the system	I can find relevant information quickly	Search bar appears with filters, search results displayed with relevant content

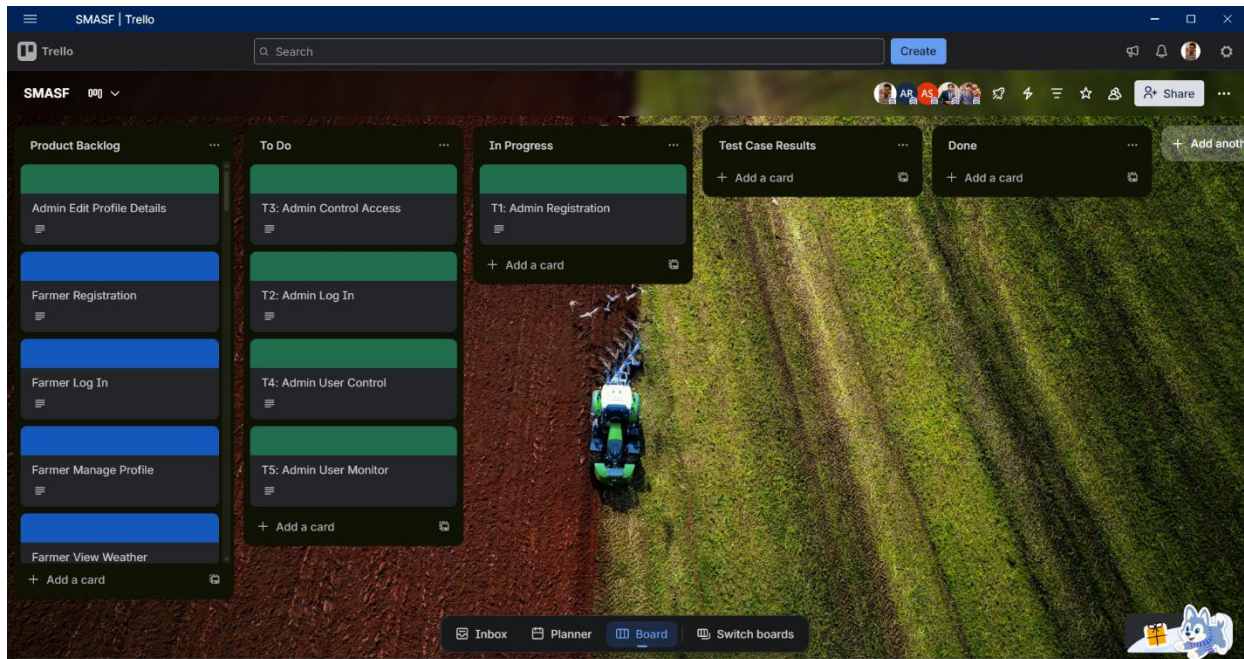


Fig 2: Shows the user story board using Trello

2.3 Requirements Traceability Matrix

2.3.1 Functional Requirements

Req. ID	Feature	Functional Requirement Description	Test Case ID	Test Case Description	Priority
1. Profile Management (All Users)					
FR-01	Registration	The system shall allow Farmers, Buyers, Experts, and Suppliers to register by entering valid personal details and selecting a role.	TC-01	Verify registration is successful with valid data.	High
FR-02	Login	The system shall allow all users to log in securely using their registered credentials.	TC-02	Verify login is successful with correct email/password.	High
FR-03	Edit Profile	The system shall allow logged-in users to update their personal information (e.g., name, phone number).	TC-03	Verify that changes to profile details are saved.	Low
2. User Management (Admin Only)					
FR-04	View Users	The Admin shall be able to view a list of all registered users in the system.	TC-04	Verify that the Admin dashboard	Medium

				displays the user list.	
FR-05	Manage Access	The Admin shall be able to remove or ban a user from the system.	TC-05	Verify that a removed user can no longer log in.	High
FR-06	Edit User	The Admin shall be able to edit user information if required.	TC-06	Verify Admin can successfully update a user's details.	Low
3. Information & Advisory (Farmer + Expert)					
FR-07	Weather	Farmers shall be able to view weather forecasts on their dashboard.	TC-07	Verify weather data loads correctly on the Farmer dashboard.	High
FR-08	Crop Tips	Farmers shall be able to read crop management tips and pest control advice.	TC-08	Verify that clicking a crop tip displays the full content.	Medium
FR-09	Expert Advice	Agricultural Experts shall be able to post best practices and answer farmer queries.	TC-09	Verify that an Expert's post appears in the feed.	Medium
FR-10	Q&A Forum	Farmers shall be able to post questions to the forum for Experts to answer.	TC-10	Verify that a Farmer can successfully post a question.	Medium

4. Market & Trade (Farmer + Buyer + Supplier)					
FR-11	List Crops	Farmers shall be able to list their produce (crops) with price and quantity.	TC-11	Verify that a new crop listing appears in the marketplace.	High
FR-12	View Price	Farmers shall be able to view current market price trends.	TC-12	Verify that market prices are displayed to the Farmer.	Low
FR-13	Browse Crops	Buyers shall be able to browse available produce and compare prices.	TC-13	Verify Buyer can view the list of all available crops.	High
FR-14	Trade Chat	Buyers shall be able to communicate directly with Farmers to negotiate deals.	TC-14	Verify that a message sent by a Buyer is received by the Farmer.	High
FR-15	Supplier Ads	Suppliers shall be able to list/advertise products (seeds, fertilizers) for sale.	TC-15	Verify that Supplier products are visible to Farmers.	Medium
5. Notifications					
FR-16	Farmer Alerts	The system shall notify Farmers about weather alerts and market changes.	TC-16	Verify Farmer receives a notification for weather updates.	Medium

FR-17	Buyer Alerts	The system shall notify Buyers when new crops are added to the system.	TC-17	Verify Buyer receives a notification when a new crop is listed.	Low
FR-18	Supplier Alerts	The system shall notify Suppliers when a Farmer places an order for their products.	TC-18	Verify Supplier receives a notification upon new order placement.	Medium

2.3.2 Non-Functional Requirements

1. Availability

- The Farmer Support App shall be available **24/7**, except during scheduled maintenance.
- The system shall ensure minimum downtime to allow farmers continuous access to important information.
- In case of temporary service failure, the system shall restore services as quickly as possible.

2. Performance

- The application shall respond to user actions within **2–3 seconds** under normal operating conditions.
- Weather updates, market prices, and notifications shall be displayed in real time or near real time.
- The system shall support multiple users accessing the platform simultaneously without significant delays.

3. Maintainability

- The system shall be designed using a modular architecture to simplify updates and bug fixes.
- Source code shall be well-documented and maintained using version control tools such as Git.

- New features or improvements can be added without affecting existing system functionality.

4. User Friendliness (Usability)

- The user interface shall be simple and easy to understand, especially for users with limited technical knowledge.
- The app shall use clear language, readable fonts, and intuitive navigation.
- Important features shall be accessible with minimal steps to improve user experience.

3. SOFTWARE DESIGN

3.1 System Design

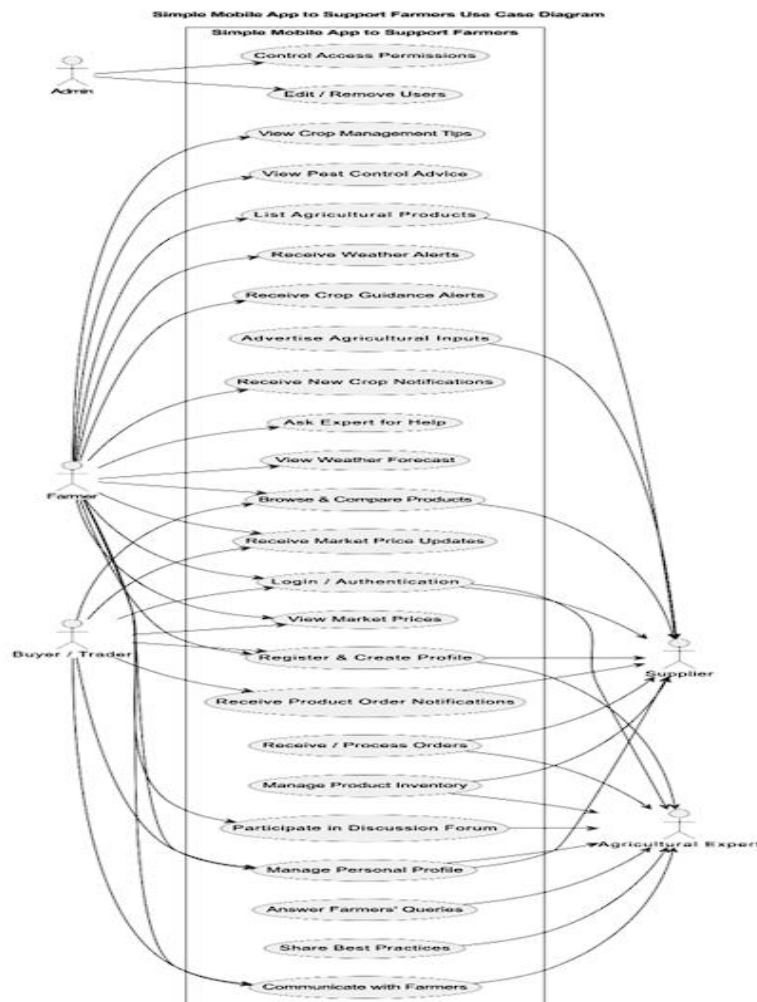


Fig 3: Shows use case diagram

The use case diagram explains a Simple Mobile App to Support Farmers that connects farmers, buyers/traders, suppliers, agricultural experts, and an admin in one integrated platform. The admin controls access permissions and manages user accounts, while farmers use the app to access crop and pest management tips, check weather and market prices, receive alerts, and browse or compare products. Buyers/traders can register, browse products, view prices, place orders, and get order notifications, whereas suppliers advertise and list products, process orders, and manage inventory. Agricultural experts support farmers by answering questions, sharing best practices, and joining discussion forums, making the app a shared digital space that improves coordination and decision-making in daily farming.

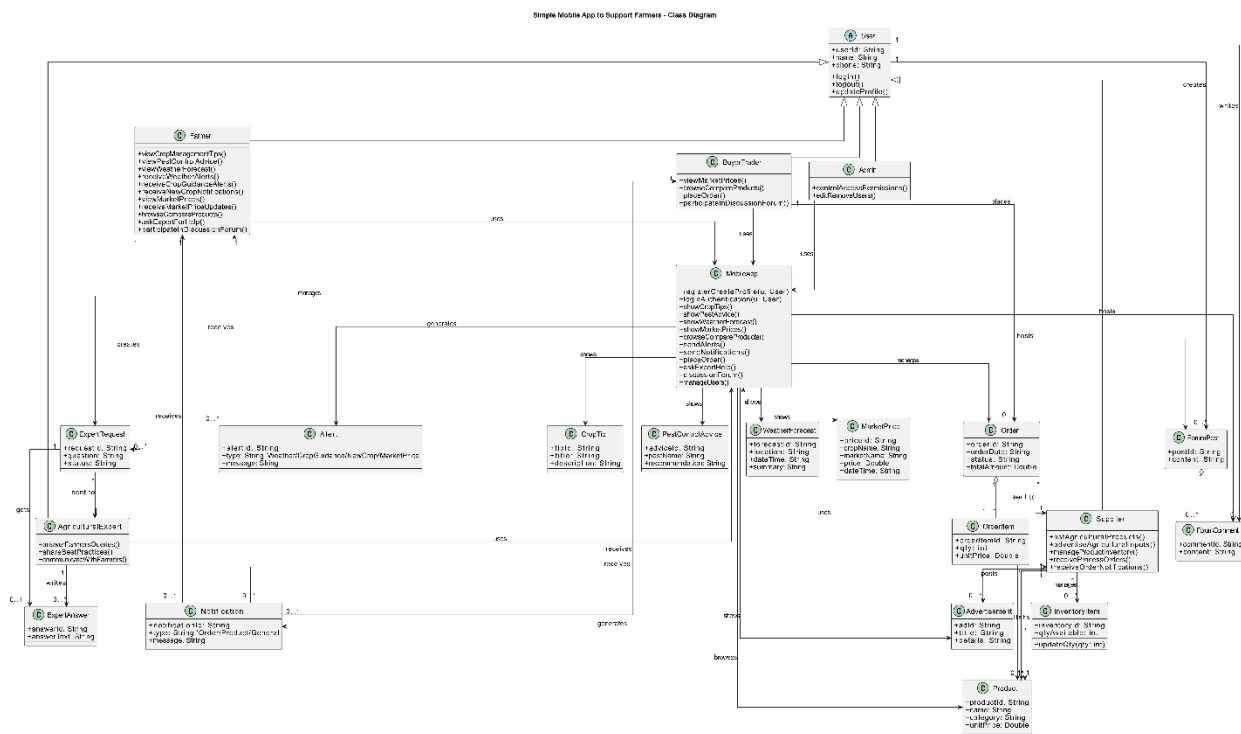


Fig 4: Shows class diagram

This class diagram represents the structure of the Simple Mobile App to Support Farmers. It shows User as the main class, connected with different roles such as Farmer, Buyer/Trader, Supplier, Agricultural Expert, and Admin. The Mobile App class controls major functions like registration, login, weather and market updates, crop guidance, and expert support. It also includes classes for marketplace activities (Product, Order, Inventory, Advertisement) and communication features (Notification, Forum Post, Expert Request, Expert Answer) to support farmers in one platform.

3.2 UI / Wireframe Design using Figma

USER LOGIN



 USERNAME:

 PASSWORD:

[Don't have an account ?Sign up](#)

LOGIN

[forgot password?](#)

Fig 5: Login page for all users.

CREATE ACCOUNT



FULL NAME :

USER NAME :

E-MAIL :

MOBILE NUMBER :

PASSWORD :

CONFIRM PASSWORD :

Back

SUBMIT

Fig 6: Farmers Registration Page.

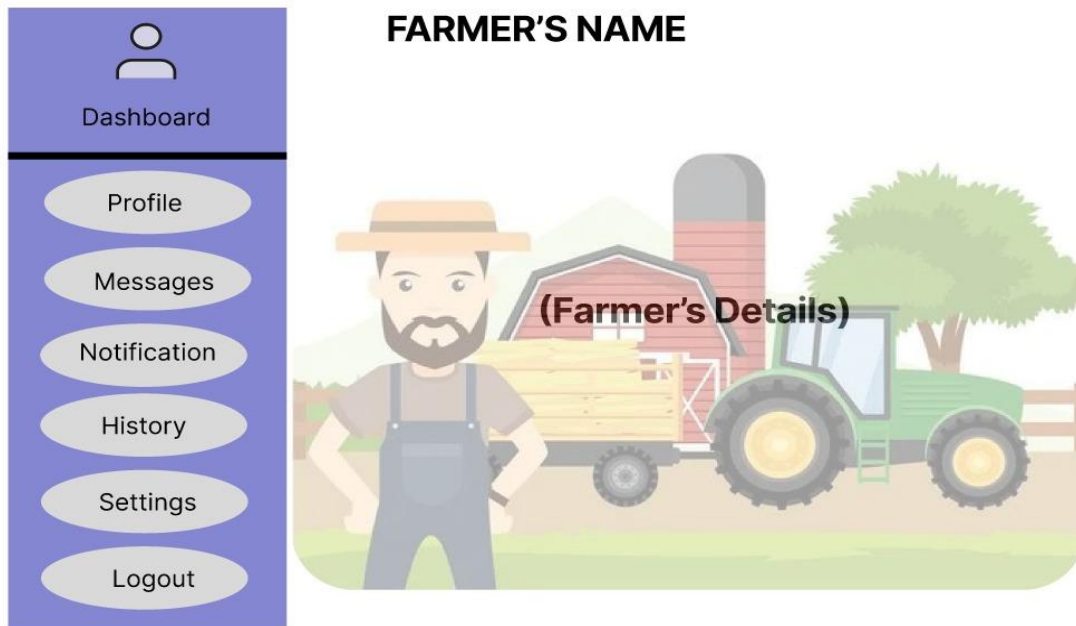


Fig 7: Farmers Dashboard Page

4. GIT WORKFLOW

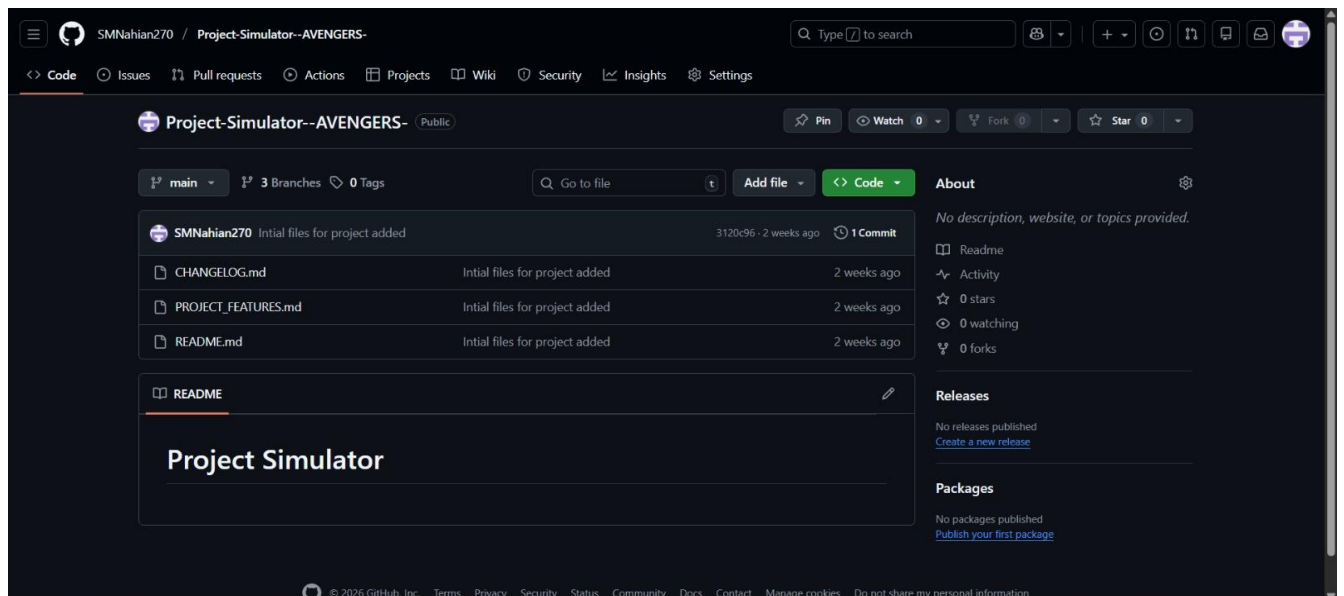


Fig 7: Git Repository after successfully 3 branches (main, dev and feat) created and 3 common .md files created and pushed to all branches through git bash.

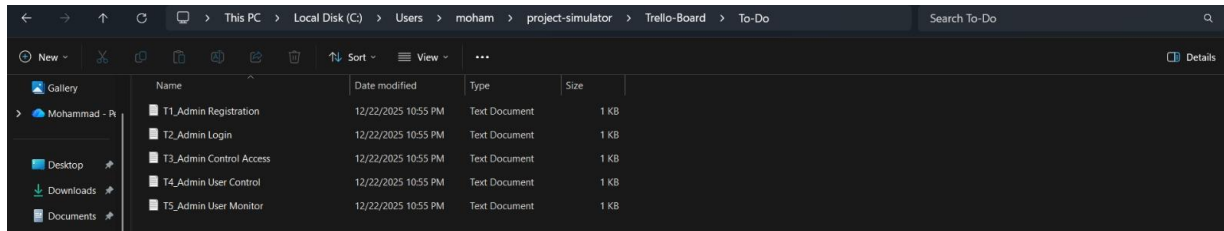


Fig 8: Local drive after successfully created two folders “Trello-Board” and “To-Do” through git bash and 5 user-story files manually to push in the git repository.

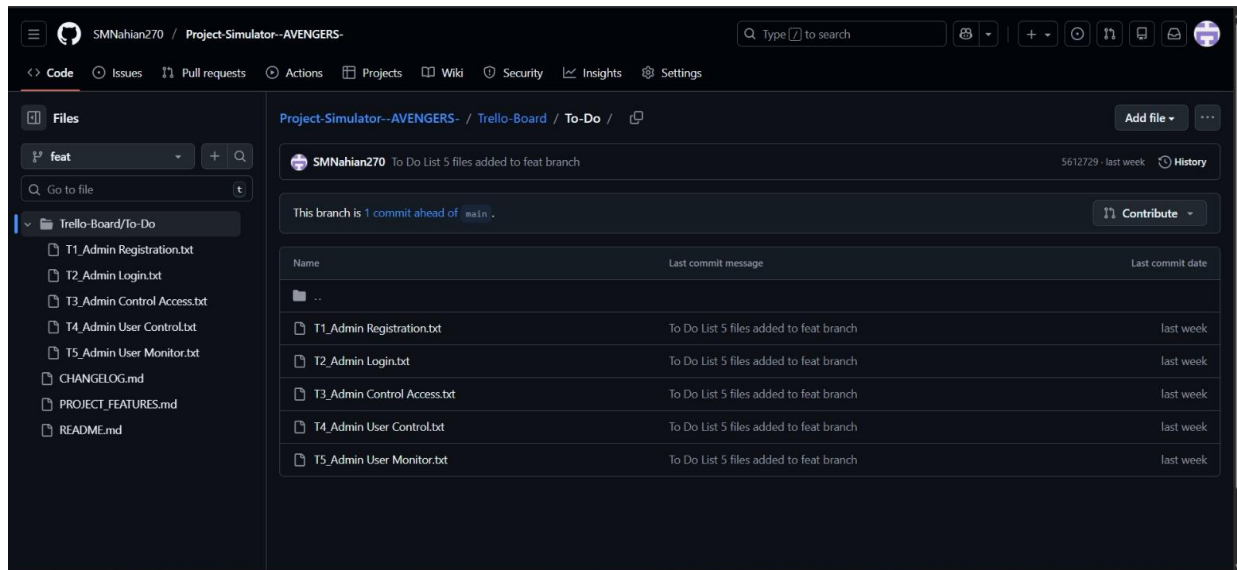


Fig 9: Git Repository after successfully 5 user story files pushed in the feat branch.

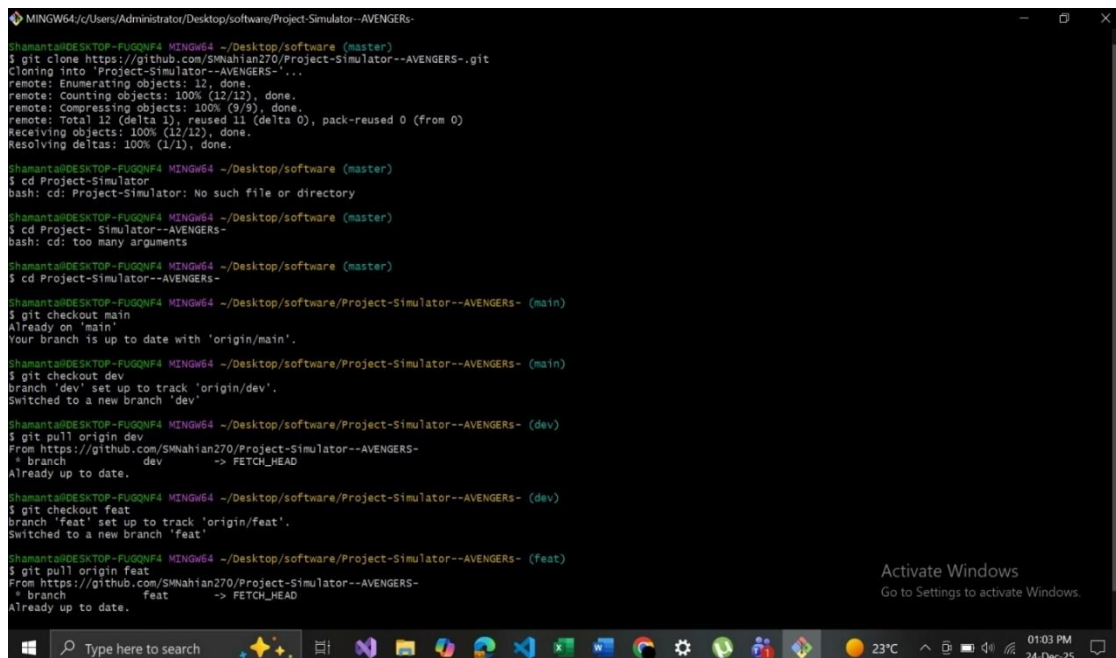


Fig 10: Git Bash commands to clone the repository and pull all branches along with files.

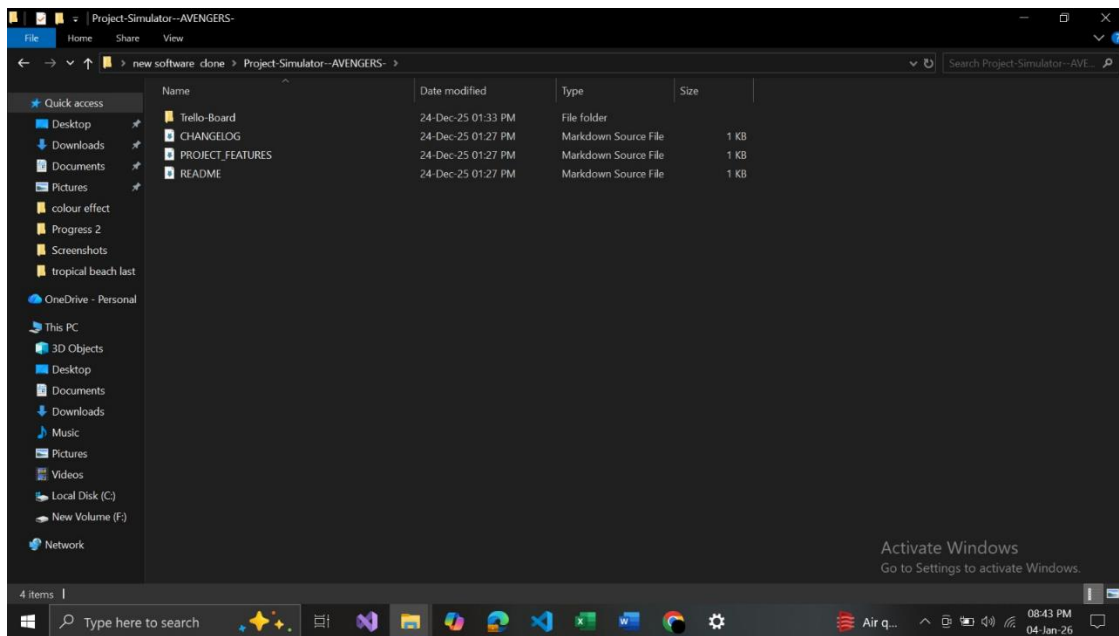


Fig 11: Local drive after successfully cloning the repository and pulling the all branches files through the git bash.

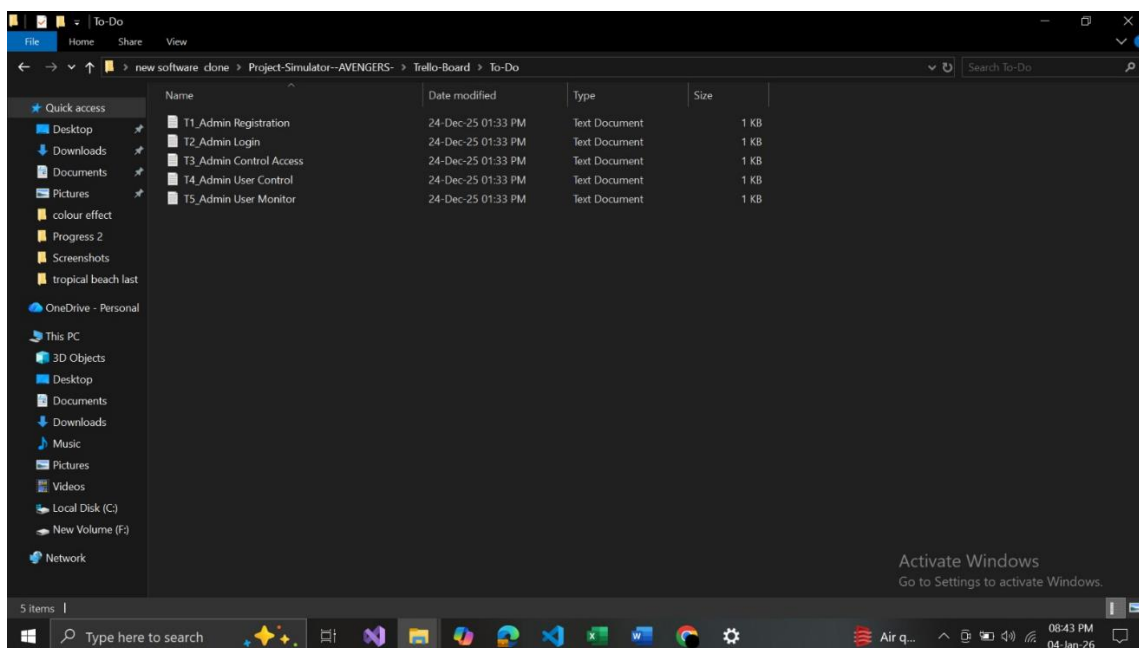


Fig 12: Local drive after successfully pulling the files from feat branch through the git bash.

5.SOFTWARE TESTING

The main goal of testing in our project Simple Mobile App for Farmers is to ensure that the application is high quality and that it meets all the requirements defined earlier.

Testing helps us detect errors before deployment, verify that each feature works as intended, and validate that the complete system supports real farmers and other users (Buyer/Trader, Supplier, Agricultural Expert, Admin). It also increases confidence that the solution is stable, secure, and ready for real-world use. In short, testing ensures both requirement conformance and software quality assurance.

For our project, we plan to use a combination of testing methods:

- **Unit Testing:**

Each module/class will be tested separately to confirm that its functions work correctly in isolation.

Example modules from our system: User, Farmer, Buyer/Trader, Supplier, Agricultural Expert, Admin, Mobile App, and core entities like Order, Order Item, Product, Inventory Item, Notification, Alert, Forum Post, Forum Comment.

- **Integration Testing:**

After individual modules are verified, they will be combined and tested together to ensure smooth interaction between components.

Example flows:

- Farmer views crop tips → receives alerts/notifications correctly
- Farmer asks expert help → expert receives request → expert sends answer back
- Buyer/Trader places an order → supplier receives the order → order notification is generated
- Forum post creation → comments added → visible to other users

- **System Testing:**

The complete system will be tested as a whole in a simulated real environment to check overall functionality, performance, and stability.

This includes end-to-end scenarios like: registration/login → browse/compare products → place order → receive notifications → use discussion forum → view weather/market updates.

- **Performance & Security Testing:**

Before release, performance and security testing will be done because the app needs fast responses and safe handling of user data.

We will focus on: expected load response time (weather/market data screens), concurrent users, login/authentication reliability, role-based access (Admin controls), input validation, and basic security checks for data protection.

- **White-Box Testing:**
White-box testing (unit tests, code-path checks, database/API query checks) will be performed during development by the developers.
It helps reduce defects early, makes later system testing smoother, and ensures internal logic, data flow, and validations are correct and efficient.

Test Cases:

Project Name: SMASF		Test Designed by: S M Nahian		
Test Case ID: T_1		Test Designed date: 31/12/2025		
Test Priority: Medium		Test Executed by: S M Nahian		
Module Name: Admin Registration Session		Test Execution date: 02/01/2026		
Test Title: Verify admin registration with valid information				
Description: Test the website admin registration page				
Precondition: Admin is on the registration page of the website				
Dependencies: no				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the site 2. Navigate to admin registration page 3. Enter username 4. Enter name 5. Enter password 6. Enter contact 7. Enter email 8. Click submit	Username: admin99 Name: Nahian Password: 321 Contact: 0177***** Email: ab@gmail.com	Admin account is successfully created. “Registration Successful” message shown.	As expected	Pass

Project Name: SMASF		Test Designed by: S M Nahian		
Test Case ID: T_2		Test Designed date: 31/12/2025		
Test Priority: Medium		Test Executed by: Name		
Module Name: Admin Login Session		Test Execution date: 02/01/2026		
Test Title: Verify Admin Login				
Description: Test the website’s login page				
Precondition: Admin has a valid username and password				
Dependencies: Admin registered successfully				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the site 2. Navigate to admin login page 3. Enter username 4. Enter password 5. Click submit	Username: admin99 Password: 321	The admin should log in to the application and redirected to admin dashboard.	As expected	Pass

Project Name: SMASF		Test Designed by: Shadman		
Test Case ID: T_3		Test Designed date: 30/12/2025		
Test Priority: Medium		Test Executed by: Shadman		
Module Name: Admin Control Access Session		Test Execution date: 02/01/2026		
Test Title: Verify Admin Control Access				
Description: Test the website’s Control Access page				
Precondition: Admin is on the Admin Dashboard Page				
Dependencies: Admin must be successfully logged into the website				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Click on Access Control Button from dashboard 2. Navigate to Control Access Page 3. Select and update any user’s access to the system	Sample User: Username: farmer123 User Status: inactive access status: active	Admin successfully updated selected user access status (to inactive), confirmation message shown.	As expected	Pass

Project Name: SMASF		Test Designed by: Shadman		
Test Case ID: T_4		Test Designed date: 30/12/2025		
Test Priority: Medium		Test Executed by: Shadman		
Module Name: Admin User Control Session		Test Execution date: 02/01/2026		
Test Title: Verify Admin User Control				
Description: Test the website’s User Control page				
Precondition: Admin is on the Admin Dashboard Page				
Dependencies: Admin Must be successfully logged into the website				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Click on User Control Button from the dashboard 2. Navigate to User Control Page 3. Select and remove any inactive user from the system	Sample User: username: seller123 User status: inactive	Admin successfully removed an inactive user from the system, confirmation message shown.	As expected	Pass

Project Name: SMASF			Test Designed by: Apurba Reza	
Test Case ID: T_5			Test Designed date: 1/1/2026	
Test Priority: Medium			Test Executed by: Apurba	
Module Name: Admin – User Monitor			Test Execution date: 03/01/2026	
Test Title: Verify Admin Can View & Monitor All Registered Users				
Description: Check if the admin can view all users and basic details.				
Precondition: Admin is Logged in.				
Dependencies: None.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Admin. 2. Navigate to “User Monitor”. 3. Click “View All Users”.	Admin account: admin01 Filter: Role = Farmer Sort: Last Active (desc)	Admin sees complete user list with pagination, filters and sorting work correctly, activity log is visible on user profile, and account management actions (Suspend/Activate, Reset Password) are available.	As expected	Pass

Project Name: SMASF		Test Designed by: MD. SADMAN ISLAM SHAMS		
Test Case ID: TC_6		Test Designed date: 01/01/2026		
Test Priority : Medium		Test Executed by: MD. SADMAN ISLAM SHAMS		
Module Name: Farmer Registration Session		Test Execution date: 03/01/2026		
Test Title: Verify farmer registration with valid information				
Description: Test the website farmer registration page				
Precondition: The farmer is on the registration page of the website and has a valid email address and contact number				
Dependencies: No				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the site 2. Navigate to farmer registration page 3. Enter username 4. Enter name 5. Enter password 6. Enter contact 7. Enter email 8. Click submit	Username: ankaramessi10 Name: Messi Password: 369 Contact: 017***** Email: ankaramessi@gmail.com	Farmer account is successfully created and a “Registration Successful” confirmation message is displayed.	As expected	Pass

Project Name: SMASF		Test Designed by: MD. SADMAN ISLAM SHAMS		
Test Case ID: TC_7		Test Designed date: 01/01/2026		
Test Priority : Medium		Test Executed by: MD. SADMAN ISLAM SHAMS		
Module Name: Farmer Login Session		Test Execution date: 03/01/2026		
Test Title: Verify Farmer Login with username & password				
Description: Test the website’s login page				
Precondition: The farmer is on the login page of the website and has a valid registered username and password				
Dependencies: The farmer account must be approved by the admin before login access is granted				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the site 2. Navigate to farmer login page 3. Enter username 4. Enter password 5. Click submit	Username: ankaramessi10 Password: 369	A “Successfully Logged In” confirmation message is displayed. The farmer is granted access to the application and is redirected to the Farmer Dashboard.	As expected	Pass

Project Name: SMASF			Test Designed by: Shamanta	
Test Case ID: T_8			Test Designed date: 31/12/2025	
Test Priority: Medium			Test Executed by: Shamanta	
Module Name: Buyer Profile Management			Test Execution date: 02/01/2026	
Test Title: Verify Buyer Profile Management				
Description: Test if the buyer can manage their profile and update personal and business information.				
Precondition: Buyer is on the Dashboard Page.				
Dependencies: Admin must grant access to the buyer to login to the system and manage the profile.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in to the app with valid id. 2. Click on "Edit Profile". 3. Change name and save.	Username: buyer123, New Name: SAM	Profile updates successfully and shows a success message.	As expected.	Pass

Project Name: SMASF			Test Designed by: Shamanta	
Test Case ID: T_9			Test Designed date: 31/12/2025	
Test Priority: Medium			Test Executed by: Shamanta	
Module Name: Buyer Agricultural Products View			Test Execution date: 02/01/2026	
Test Title: Verify Buyer Can View Agricultural Products				
Description: Test if the buyer can view available agricultural products listed by farmers.				
Precondition: Buyer is currently viewing the Dashboard.				
Dependencies: Admin must grant the buyer access to the buyer to login to the system and view the agricultural products.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in to the app. 2. Click on "Marketplace". 3. Click on a vegetable item.	Product: Fresh Potato, Price: \$10	Buyer successfully sees the product list and item details.	As expected.	Pass

Project Name: SMASF			Test Designed by: Apurba	
Test Case ID: T_10			Test Designed date: 1/1/2026	
Test Priority: Medium			Test Executed by: Apurba	
Module Name: Farmer View Market Price			Test Execution date: 03/01/2026	
Test Title: Verify Farmer Can View Current Market Prices				
Description: Check if the farmer can view current market prices for crops.				
Precondition: Farmer is logged in.				
Dependencies: Market price data available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app and navigate to “Market”. 2. Select crop from list (e.g., Rice). 3. View current market price and open historical chart.	Crop: Rice; Market: Dhaka; Date: 3 Jan 2026	Farmer successfully sees current price and can view historical trend.	As expected	Pass

6. CONCLUSION

In conclusion, the Farmer Support App is designed to provide a practical and user-friendly digital solution for farmers by bringing essential agricultural services into a single platform. Throughout this project, we focused on identifying real-life problems faced by farmers, such as lack of timely information, limited market access, and dependence on middlemen. By applying software engineering concepts like requirement analysis, system design, version control, and testing, we were able to plan a structured and effective solution.

This project helped us understand how software can play an important role in solving real-world problems. Features such as weather updates, crop management guidance, and direct buyer–seller communication can help farmers make better decisions, reduce losses, and improve their income. At the same time, buyers, experts, and suppliers benefit from transparent and efficient communication.

From an academic perspective, this project strengthened our understanding of the software development lifecycle, teamwork, and practical implementation of theoretical knowledge. Overall, the Farmer Support App demonstrates how technology can support sustainable agriculture and improve rural livelihoods. With further development and real-world deployment, this system has the potential to make a positive impact on the agricultural sector.